

Case / Scan ID:	LCT-85725	Patient Case:	ANONYMOUS (Clinical Study)
File Name:	test_ct_slice.png	Analysis Model:	Attention U-Net Segmentation (Poisson + Speckle)
Modality:	Computed Tomography (CT)	Report Status:	COMPLETED (AI-Generated)

CT Scan Imagery & Diagnostic Overlay

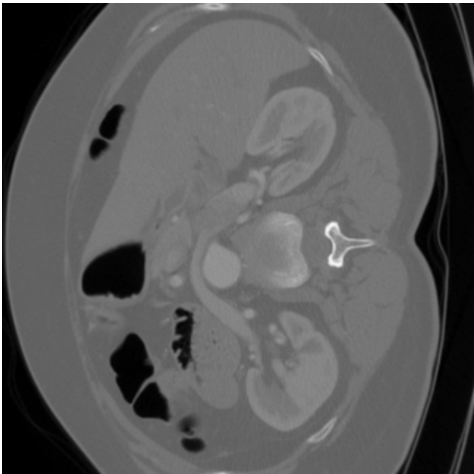


Figure 1: Original CT Scan
Input diagnostic image

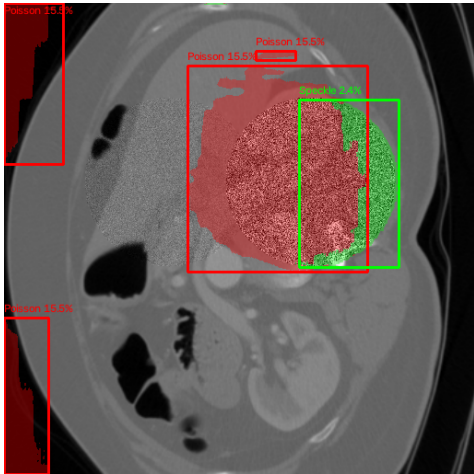


Figure 2: Annotated CT Scan (Red: Poisson | Green: Speckle)

Quantitative Noise Classification & Metrics (Model 2 (Attention U-Net))

Noise Classification	Pixel Count / Prob.	Severity / Score	Clinical Classification
Poisson Noise (Photon Shot)	0 px	15.47%	CRITICAL
Speckle Noise (Multiplicative)	0 px	2.40%	MODERATE
Clean Signal Region	215,291 px	82.13%	Normal / Target
TOTAL NOISE (Model 2)	0 px	17.87%	CRITICAL

Clinical Interpretation & Denoising Recommendations

Model Assessment (Model 2 (Attention U-Net)): Elevated noise artifacts detected (17.9%). Structural detail is noticeably compromised by noise patterns.
Radiologist Action: Strongly recommend executing deep neural restoration or wavelet-based denoising. If diagnostic ambiguities persist, evaluate sensor hardware.

Radiologist Review Block	AI System Validation Specs
Reviewing Radiologist Signature	Architecture: Attention U-Net (Joshna.pth) Gating: Additive Attention Gates (AG) Classes: Poisson, Speckle Noise

Disclaimer: This report was automatically generated by Model 2 (Attention U-Net) for multi-noise classification and severity assessment in educational & research environments. Findings should be cross-referenced by a qualified medical professional prior to clinical intervention.